3D Medical Visualization

Submitted in partial fulfilment of the requirements of the degree of

BACHELOR OF ENGINEERING

by

Kunal Redij: 23106067

Makarand Panchal:23106065

Rushikesh Shinde:23106061

Guide:
Dr. Tejashri Kolhe

Department of Computer Science & Engineering
(Artificial Intelligence & Machine Learning)

A. P. SHAH INSTITUTE OF TECHNOLOGY,THANE

UNIVERSITY OF MUMBAI

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A. P. SHAH INSTITUTE OF TECHNOLOGY, THANE

CERTIFICATE

This is to certify that the project entitled "3D Medical Visualization Tool" is a Bonafide work of **Kunal Redij (23106067) , Makarand Panchal (23106065), Rishikesh Shinde (23106061)** submitted to the University of Mumbai in partial fulfilment of the requirement for the award of the degree of **Bachelor of Engineering** in **Computer Science & Engineering (AI&ML)**

Dr.Tejashri Kolhe Project Guide	Prof. Yogeshwari Hardas Project Co-Ordinator
Dr. Jaya Gupta Head of Department	Dr. Uttam D Kolekar Principal

A. P. SHAH INSTITUTE OF TECHNOLOGY, THANE

Project Report Approval for T. E.

This project report entitled ***3D Medical Visualization*** by ***Kunal Redij,Makarand Panchal,Rishikesh Shinde*** is approved for the degree of ***Bachelor of Engineering*** in **Computer Science & Engineering (Artificial Intelligence & Machine Learning)**, ***2025-26***.

Examiner Name	Signature
1._____	_____
2. _____	_____

Date:

Place:

Declaration

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Kunal Redij
23106067

Makarand Panchal
23106065

Rishikesh Shinde
23106061

Date:

Abstract

In the field of medical diagnostics, the analysis of Computed Tomography (CT) scans for pulmonary abnormalities is a cornerstone of modern healthcare. However, this process traditionally relies on specialized, desktop-bound software that can be costly and inaccessible, creating a barrier for rapid, intuitive visualization and collaborative review. As medical data volume grows, there is an increasing need for accessible, integrated systems capable of streamlining the workflow from raw scan data to actionable visual insight. This project, titled "3D Med: A CT Scan Visualization Platform," addresses these challenges through a comprehensive web-based solution that automates the analysis and interactive 3D visualization of pulmonary CT scans using advanced image processing and web rendering technologies.

By improving accessibility, reducing cost barriers, and enabling cloud-based collaborative visualization, **3D Med directly contributes to SDG 3 (Good Health and Well-Being)** by supporting early detection and better management of respiratory diseases. Moreover, through the integration of open-source technologies and scalable web infrastructure, the project aligns with **SDG 9 (Industry, Innovation and Infrastructure)** by demonstrating how digital innovation can strengthen healthcare systems and democratize access to diagnostic tools. The successful implementation of 3D Med highlights the transformative potential of combining medical image processing with modern web technologies to create sustainable, inclusive, and impactful diagnostic solutions.

Keywords: Computed Tomography (CT), 3D Visualization, Medical Imaging, Flask, SimpleITK, PyVista, three.js, GLTF, Lung Segmentation, Anomaly Detection, Image Processing, Validation Metrics, Web-Based Platform.

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Chapter 1

Introduction

The rapid growth of medical imaging technologies has significantly improved diagnostic accuracy and clinical decision-making. Among these modalities, Computed Tomography (CT) scans are particularly crucial for detecting and monitoring lung-related diseases such as cancer, pneumonia, and COVID-19. However, manual interpretation of CT scans by radiologists is a challenging task it is time-intensive, subject to human fatigue, and prone to inter-observer variability. With the exponential increase in patient data, there is a pressing need for intelligent systems that can assist in both analysis and visualization of CT scans.

To address this challenge, our project "3D Med" is developed as a web-based system that combines deep learning–based medical image analysis with interactive 3D visualization. The core idea is not just to classify or detect abnormalities in CT scans, but also to present results in a visually interpretable form that empowers clinicians and students alike to understand model predictions. The system leverages convolutional neural networks (CNNs) for feature extraction and abnormality detection, while visualization layers (such as heatmaps and 3D volume rendering) ensure that outputs are transparent, interpretable, and clinically useful.

Unlike conventional black-box AI models, 3D Med emphasizes explainability by highlighting suspicious regions in CT scans where abnormalities may exist. This helps radiologists validate

the AI's decision rather than blindly relying on it. Additionally, the platform's web-based interface ensures accessibility, scalability, and ease of integration into healthcare workflows.

By merging automated analysis with interactive visualization, 3D Med contributes to faster diagnosis, reduced workload for medical professionals, and enhanced trust in AI-assisted healthcare. Furthermore, it serves as a valuable educational tool for medical students, allowing them to explore 3D CT visualizations alongside AI-driven insights

Chapter 2

Literature Survey

2.1 History

Medical image visualization has always been an essential complement to diagnosis. In traditional radiology, CT scans are manually examined slice by slice, which is time-consuming and subject to human variability. To assist radiologists, early computational tools (1990s– 2000s) focused on 2D rendering of slices with basic contrast adjustments and threshold-based highlighting of suspicious regions. These were useful but limited in capturing 3D anatomical context.

With the advancement of 3D reconstruction techniques (2000s–2010s), volumetric visualization became possible, enabling radiologists to rotate and inspect CT scans in multiple planes (axial, sagittal, coronal). Parallelly, Computer-Aided Diagnosis (CAD) systems began integrating visualization overlays such as bounding boxes and segmentation masks to highlight nodules or lesions.

The deep learning era (2010s–present) introduced explainable AI and advanced visualization techniques. Convolutional Neural Networks (CNNs) achieved high accuracy in lung nodule detection and infection classification, while visualization methods like Grad-CAM (Gradient weighted Class Activation Mapping), saliency maps , and heatmaps allowed systems to Highlight regions influencing predictions . This bridged the gap between “ black-box ” AI

and human interpretability. Today, interactive dashboards and web-based visualization tools provide radiologists and clinicians with dynamic ways to inspect CT scans, validate AI predictions, and support decision-making.

2.2 Literature Review

[1] Ardila et al., Nature Medicine, 2019 – *End-to-end lung cancer screening with deep*
This landmark work by Google Health used deep learning for early lung cancer detection. Importantly, it integrated 3D volumetric visualization, allowing radiologists to interact with AI predictions and review highlighted suspicious regions. The visual overlays improved radiologist trust and reduced false positives.

[2]Wangetal.,2020–COVID-Net
COVID-Net introduced a tailored CNN for detecting COVID-19 from chest imaging. A key contribution was its explainability: predictions were accompanied by Grad-CAM heatmaps highlighting infected lung regions. This ensured clinicians could see where the model focused, bridging trust between AI and medical experts.

[3] LITJENS et al., 2017 – Deep learning in medical image analysis (Survey)
A broad survey showing how visualization evolved from traditional rendering to AI-guided overlays. It emphasized that effective visualization is crucial not just for radiologist trust, but also for clinical validation and regulatory approval.

[4]Selvarajuetal.,2017-Grad-CAMframework
While not specific to CT scans, Grad-CAM revolutionized visualization in CNN-based medical imaging. Its heatmaps are widely used in CT and MRI analysis to visually link AI predictions to specific regions of interest. Many CT analysis systems today integrate Grad-CAM or similar methods in their visualization layer.

[5] Hussain et al., 2021 - A survey on COVID-19 detection from medical images
The authors reviewed CT/X-ray deep learning models and emphasized visualization as a critical factor for clinical adoption. They highlighted that AI predictions without visual proof are rarely accepted in clinical practice, making interpretability tools indispensable.

2.3 Research Gap & Motivation

- Beyond classification: Many studies report high classification accuracy but lack intuitive visualization tools for clinicians.
- Explainability: AI remains a “black box” without visualization overlays like heatmaps, saliency, or segmentation masks.
- User-centered dashboards: Few systems provide interactive, web-based platforms that allow doctors to toggle slices, zoom into regions, or compare AI predictions with original CT scans.
- Integration challenge: Current research often focuses either on accuracy or visualization, but not both in a unified workflow.

2.4 Summary

Visualization plays a dual role: it enhances interpretability of AI models and supports clinical decision-making. Historical methods relied on static CT rendering, but modern systems incorporate deep learning with heatmaps, 3D reconstructions, and interactive dashboards. Literature shows strong progress in combining detection with visualization, but there remains a need for integrated, real-time, web-based solutions where radiologists can inspect scans, validate AI predictions, and trust the decision pipeline.

Chapter 3

Limitations in Existing Systems

Despite significant progress in medical image analysis and visualization, current systems face several limitations that restrict their widespread clinical adoption:

1. Black-Box Nature of AI Models
   - Many deep learning systems achieve high accuracy in detecting lung diseases but fail to explain why a decision was made.
   - Without visualization tools such as heatmaps or segmentation masks, radiologists cannot fully trust or validate the AI's predictions.
2. Limited Visualization Capabilities
   - Traditional systems often display CT scans as static 2D slices, which do not capture the full 3D structure of the lungs.
   - Few frameworks provide interactive, volumetric visualization where doctors can rotate, zoom, or explore infected regions in 3D.
3. Over-Reliance on Classification
   - Many existing approaches focus only on classification (e.g., COVID-positive or negative) without providing supporting evidence or highlighting affected areas.
   - This makes them unsuitable for detailed medical decision-making, where localization and severity analysis are equally important.
   - Lack of interpretability in such models reduces clinician trust, as they cannot visualize or understand the reasoning behind the system's predictions.

4. Data Generalization Issues
    - AI models often perform well on curated datasets but show reduced accuracy when tested on real-world, diverse hospital data.
    - Differences in scanner quality, resolution, and patient demographics reduce generalization capability.
5. Integration Challenges
    - Most research prototypes are built as isolated models and lack user-friendly interfaces.
    - Radiologists need web-based, accessible systems that can seamlessly fit into existing healthcare workflows, but many current tools remain difficult to deploy or use in real-time.
6. Lack of Educational Utility
    - Existing systems primarily target diagnosis but rarely serve as learning tools for medical students.
    - There is limited support for interactive visualization that helps students understand anatomy, pathology, and AI reasoning together.

Table 3.1: Summary of limitations of existing systems

Title	**Conference / Journal Details**	**Key Points**	**Improvements Proposed**	**Citation**
End-to-EndLung CancerScreening with Deep Learning on Low-Dose CT	*Nature Medicine*, 2024 (Google Health)	2024 (Google Health) 3D CNN model detects lung cancer with radiologist-level accuracy; integrated volumetric visualization	Improve explainability and reduce true & false positives through interpretable overlays	[1]

COVID-Net: A Tailored & Deep Convolutional Neural Network for COVID-19 Detection	*arXiv / IEEE Access*, 2021	Custom CNN for COVID-19 for detection from CT/X-ray; includes Grad-CAM heatmaps for interpretability	Enhance robustness on diverse datasets, expand to multi-disease detection	[2]
Deep Learning in Medical & Image Analysis – A Survey	*Medical Image Analysis*, 2021	Comprehensive review of CNN, RNN, autoencoders for CT, MRI, X-ray;emphasizes role of visualization	Develop in hybrid models combining accuracy with real-time3D visualization	[3]
Grad-CAM: Visual Explanations from Deep Networks	*ICCV*, 2019	Introduced heatmaps for model interpretability; widely adopted in medical imaging	Adapt Grad-CAM for volumetric (3D CT) visualization, improve localization	[4]
A Comprehensive Survey on COVID-19 Detection Using CT and X-ray Images	*Elsevier Computers in Biology and Medicine*, 2021	Survey of AI approaches; accuracy 85–95% using CNNs; stresses importance of visual proof for acceptance	Create interactive dashboards that combine AI predictions with user-centered visualization	[5]
nnU-Net: A Self-Adapting Framework for Medical in Image Segmentation	*Nature Methods*, 2021	Automated segmentation pipeline; effective for CT lesion/nodule visualization	Integrate segmentation masks with classification results in real-time visualization systems	[6]

Chapter 4

Problem Statement & Objectives

4.1 Problem Statement

Pulmonary diseases such as lung cancer and infections represent major global health challenges, necessitating precise and timely diagnosis. Computed Tomography (CT) scans stand as one of the most effective imaging modalities for detecting lung abnormalities. However, the complexity of lung anatomy combined with subtle pathological changes makes manual interpretation of CT images by radiologists a difficult and time-consuming task. Traditional analysis relies heavily on reviewing two-dimensional slices, which limits spatial understanding and may result in missed nodules or diagnostic errors. Furthermore, the increasing volume of scans in clinical practice creates workload pressures that can compromise diagnostic quality.

While automated and AI-assisted imaging techniques have been developed to augment radiologist capabilities, current systems often fall short due to limited accuracy, lack of robust validation, poor visualization options, and inadequate workflow integration. Many existing solutions do not support interactive 3D exploration to thoroughly assess nodules and their spatial context. Additionally, the absence of comprehensive reporting and scan management tools inhibits longitudinal patient monitoring and multidisciplinary collaboration. These gaps signify an urgent need for an integrated, reliable, and user-friendly platform that combines advanced AI detection with validated results and rich visualizations to enhance clinical decision-making in pulmonary CT scan analysis.

4.2 Objectives

In order to address the challenges identified in the problem statement related to the manual, time-consuming, and often inconsistent interpretation of pulmonary CT scans, this project sets out the following objectives:

1. **To automate pulmonary CT scan analysis:** Develop an AI-powered platform capable of automatically segmenting lung tissues and detecting nodules and other anomalies in volumetric CT data, thereby reducing the dependency on manual slice-by-slice review by radiologists.
2. **To enhance image quality and interpretation accuracy:** Implement advanced image pre-processing techniques, including contrast enhancement and noise reduction, to improve the clarity and diagnostic value of CT scans, enabling robust detection performance even in challenging cases.
3. **To incorporate validation against expert annotations:** Enable the system to compare automated detections with ground truth data from clinical annotations, providing metrics such as sensitivity, specificity, and confidence scores that support transparent and trustworthy diagnostic outcomes.
4. **To deliver interactive 3D visualization:** Integrate dynamic, web-based 3D rendering of lung structures and detected anomalies, allowing clinicians to intuitively explore spatial relationships, which is critical for comprehensive assessment and clinical decision making.
5. **To streamline clinical workflows:** Facilitate efficient scan management by creating a centralized library with easy retrieval, review, and deletion functionalities, improving case tracking, longitudinal studies, and audit processes across institutions.
6. **To provide customizable analysis options:** Allow users to adjust parameters such as scan resolution, gamma correction, and color mappings to tailor the processing pipeline and visualization outputs according to specific clinical requirements and preferences.
7. **To ensure system scalability and accessibility:** Build the platform on modular architecture with open web standards, enabling deployment across diverse environments from standalone workstations to cloud-based services and access from standard web browsers without the need for specialized software installations.
8. **To foster future extensibility:** Architect the system with modular components and flexible interfaces, permitting easy incorporation of new AI models, support for additional imaging modalities, and integration with hospital information systems to future-proof the platform

Chapter 5

Proposed System

The proposed system, 3D Med, is designed as an intelligent CT scan analysis and visualization platform. It processes raw CT scan data, applies deep learning–based analysis to detect potential abnormalities, and then presents the results through rich and interactive 3D visualizations. The system ensures that scans are preprocessed to enhance quality and consistency, after which abnormalities such as nodules or lesions are automatically identified. Along with detection, quantitative information such as tissue volumes and anomaly measurements is extracted, providing valuable insights for diagnosis.

A unique feature of 3D Med is its emphasis on visualization. Instead of offering only classification results, the system renders the CT data in an interactive 3D environment where suspicious regions are highlighted. This not only makes the analysis results more interpretable for doctors but also provides an educational platform for medical students to explore lung anatomy and disease patterns in detail. By combining automated analysis with clear visualization, the system bridges the gap between artificial intelligence outputs and clinical decision-making.

Why It Works

3D Med works effectively because it integrates three essential elements: accurate analysis, explainable outputs, and interactive visualization. Preprocessing and anomaly detection ensure that the system identifies meaningful medical patterns, while 3D visualization makes these findings transparent and easy to validate. This combination transforms raw scan data into actionable insights that radiologists can trust and students can learn from. By aligning advanced

deep learning techniques with human-centered visualization, the system achieves both accuracy and usability, making it suitable for real-world healthcare and academic environments.

5.1 System Architecture

The architecture of the proposed system, 3D Med, is designed as a multi-stage modular pipeline, transforming raw CT scan data into actionable diagnostic insights and interactive 3D visualizations. As illustrated in Figure 5.1, the workflow begins at the input stage, where CT scan files are uploaded through the browser interface. The uploaded data is passed to the backend via defined API endpoints, processed through the CT analysis pipeline, and then returned to the user as quantitative metrics and 3D visual outputs.

The design ensures that each stage of the pipeline contributes to data refinement, anomaly detection, and visualization. The system is not limited to simple classification but instead emphasizes localization, segmentation, and interpretability, thereby addressing the limitations of traditional black-box AI approaches. The modular structure also ensures extensibility, allowing for future improvements such as multi-disease detection, uncertainty estimation, and clinical integration.

Moreover, the 3D Med architecture is built for scalability and efficiency, ensuring smooth handling of large CT datasets. Advanced preprocessing improves image quality for accurate segmentation and analysis. The visualization module enables real-time 3D exploration, offering clinicians clear anatomical insights. Integrated monitoring ensures stable performance and reliability.

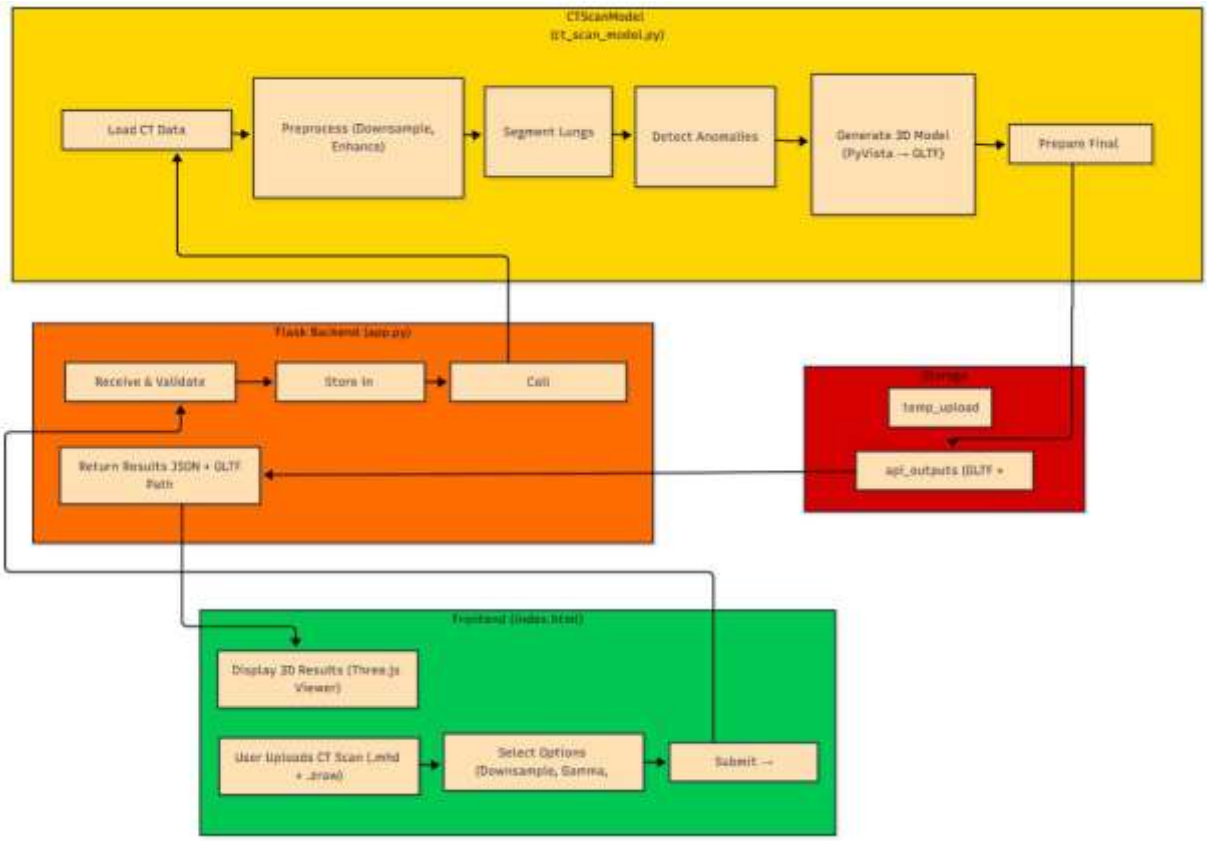

Figure 5.1: Overall system architecture & workflow

Description of figure 5.1:

- **InputStage(Browser)**:

  Users interact with the system through a web interface (index.html), where CT scans in formats such as .mhd and .zraw are uploaded. The interface integrates Three.js with a GLTF viewer, enabling interactive exploration of CT volumes.

- **FlaskAPI(app.py):**

  The backend, implemented in Flask, handles API routes such as /api/analyze, /api/status, /api/scans, and /api/cleanup. It manages CT file uploads, initiates model analysis, and retrieves processed outputs.

- **CTScanModel(ct_scan_model.py):**

  This is the core analysis engine responsible for prediction. It performs:

  - Pre-processing: Loading CT volumes via SimpleITK, resampling to consistent voxel spacing, segmentation of lung regions, and contrast enhancement.
  - Analysis: Detects anomalies, compares annotations (when available), and evaluates outputs (TP/FP/FN).
  - Visualization Preparation: Converts results into GLTF-compatible outputs and embeds anomaly markers for rendering.

- **Storage**:

  Intermediate and final results are stored in structured directories (temp_uploads for raw scans, api_outputs for processed data such as GLTF models and logs).

- **OutputStage(Browser_Results):**

  Results are returned to the browser as JSON metrics, GLTF models, and scan libraries. The user can interactively view the CT scans with highlighted anomalies, rotate or zoom the volume, and inspect quantitative measures such as tissue volumes or detected nodules.

5.2 System Modules

Input & Upload Module

- Accepts CT scan volumes (.mhd, .zraw).
- Provides a user-friendly upload interface through the browser (Three.js + GLTF viewer).

Pre-processing Module

- Uses *SimpleITK* to load and normalize scan volumes.
- Performs resampling to isotropic voxel spacing.
- Applies segmentation to isolate lung regions and reduce background noise.
- Enhances contrast for better anomaly detection.

Analysis Module

- Detects anomalies such as nodules or lesions using algorithmic/ML-based methods.
- Matches annotations (when present) to validate accuracy.
- Computes evaluation metrics (True Positives, False Positives, False Negatives).

Visualization Module

- Exports CT volumes and results into *GLTF format* for browser rendering.
- Adds anomaly markers, lesion contours, and 3D overlays.
- Enables interactive rotation, zooming, and slice navigation.

Storage & Retrieval Module

- Stores raw uploads in temp_uploads.
- Saves processed outputs (GLTF models, metrics, logs) in api_outputs.
- Ensures results can be reloaded without reprocessing.

Result Presentation Module

- Displays processed results on the browser as:
- *JSON metrics* (volumetrics, anomaly counts).
- *3D GLTF visualization* (interactive exploration).
- *Scan library view* (organized archive of analyzed scans).

5.3 Workflow Explanation

The workflow of 3D Med begins when the user uploads a CT scan through the browser interface. The uploaded data is sent to the Flask API, which stores it in the temporary directory and triggers the analysis pipeline within ct_scan_model.py. The pre-processing stage ensures uniformity in resolution and contrast, preparing the scan for robust analysis. The analysis stage then detects potential anomalies and calculates performance metrics if reference annotations are available.

Following analysis, the visualization module exports the results into a GLTF-compatible format along with anomaly markers. These outputs are stored and then delivered back to the browser, where the user can view both quantitative metrics (JSON reports) and qualitative results (3D interactive visualization). The workflow ensures that the final outputs are not only accurate but also transparent and interpretable, making the system suitable for medical professionals and educational use.

As illustrated in Figure 5.1, the workflow follows a clearly modular and interconnected structure, where each component performs a specific role in the data processing pipeline. The frontend (green section) manages user interaction, file upload, and visualization, while the Flask backend (orange section) serves as the communication bridge between the interface and the analysis model. The CTScanModel (yellow section) represents the computational core, executing preprocessing, segmentation, and anomaly detection before generating the final 3D model. Supporting this workflow, the storage module (red section) efficiently handles temporary and output files, ensuring smooth data transfer between stages. This organized flow enhances maintainability, scalability, and transparency within the entire system.

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Chapter 6

Experimental Setup

6.1 Frontend Setup

The frontend of 3D Med serves as the user interface through which medical professionals and students interact with the system. It was developed using standard web technologies with Three.js for rendering volumetric CT data and GLTF models in the browser.

Technologies Used:

- **HTML5 & CSS3:** for structuring and styling the visualization interface.
- **JavaScript (ES6):** for dynamic user interaction and event handling.
- **Three.js:** for 3D rendering of CT volumes and anomaly overlays.
- **GLTF Viewer:** for efficient visualization of 3D models exported by the backend.

UI Features:

- Simple and user-friendly design for uploading CT scan files (mhd + zraw).
- 3D interactive viewer with rotation, zoom, and panning features.
- Visualization of anomaly markers and segmented regions in real time.
- Scan library for managing previously uploaded scans.
- Responsive layout compatible with desktops and modern browsers.

6.2 Backend Setup

The backend forms the core logic of the 3D Med system, handling CT scan ingestion, pre-processing, anomaly detection, and preparation of visualization-ready outputs. It was implemented using Flask for API development and integrates with the CT scan model (ct_scan_model.py) for analysis.

Technologies Used:

- **Flask (Python):** for creating RESTful APIs and managing routes.
- **Python 3.10+:** as the core programming language.
- **SimpleITK:** for CT scan loading, resampling, and preprocessing.
- **NumPy & SciPy:** for mathematical computations and array operations.
- **PyVista (optional):** for generating 3D surface models of CT scans**.**
- **Pandas:** for structured handling of anomaly metrics and reports.

Backend Functional:

- Handles CT file uploads (via /api/ analyze and related endpoints).
- Performs pre-processing: normalization, segmentation, and contrast enhancement.
- Conducts analysis: anomaly detection, nodule annotation, and evaluation metrics (e.g., TP/FP/FN).
- Exports outputs in GLTF format for visualization and JSON format for metrics.
- Provides scan management through structured storage (temp_uploads/, api_outputs/).
- Implements secure file handling and validation to ensure data integrity and prevent corrupted or unauthorized uploads.
- Uses modular APIs for seamless communication between frontend, model pipeline, and storage components.
- Supports asynchronous processing to handle multiple user requests efficiently without affecting performance.
- Logs all activities and processing steps for transparency, debugging, and audit purposes.
- Facilitates easy scalability and integration with future modules such as multi-disease detection or cloud deployment.
- Implements configurable thresholds for anomaly detection to adapt to different diagnostic requirements.
- Includes unit and integration testing frameworks to ensure backend reliability and consistent performance across updates
- Utilizes error-handling mechanisms to manage failed uploads, incomplete scans, or corrupted data gracefully.

6.3 Development Environment:

The development environment was configured to ensure smooth integration and testing of all modules.

Environment Details:

IDE: Visual Studio Code

API Testing Tools: Postman / REST Client

Version Control: Git (optional for project tracking)

Operating System: Windows 10 / 11 (64-bit)

The experimental setup of 3D Med provides a robust foundation for CT scan analysis and visualization. The frontend, powered by Three.js and GLTF rendering, ensures an intuitive and responsive interface for exploring medical scans in 3D. The backend, developed using Flask and Python scientific libraries, performs preprocessing, anomaly detection, and result generation with accuracy and scalability. A well-configured development environment with the necessary software libraries and hardware support enabled smooth integration and real-time performance, successfully validating the effectiveness of the proposed architecture.

Chapter 7
Results & Discussion

7.1 Results

The implemented CT Scan Analysis and Visualization System successfully achieved accurate detection and visualization of lung regions and possible anomalies from volumetric medical imaging data. The backend model effectively process ".mhd and .zraw" CT scan files using SimpleITK, performing essential preprocessing steps such as downsampling, contrast enhancement, and gamma correction. Through segmentation and masking algorithms, the system isolates lung tissues from other anatomical structures, enabling precise anomaly detection. The algorithm identifies regions of interest based on intensity thresholds and spatial parameters, providing reliable detection outputs along with detailed quantitative information such as voxel spacing, tissue volume, and anomaly confidence levels.

The results are presented through a responsive web-based interface that allows users to upload CT scans, select preprocessing parameters, and visualize the analysis outcome interactively. The 3D reconstruction of lung models, generated using PyVista and rendered via Three.js, provides a clear and intuitive representation of normal and anomalous regions. The system outputs both JSON summaries for computational insights and GLTF 3D models for visual interpretation. Overall, the project demonstrates an efficient and complete workflow from data extraction and processing to visualization highlighting the potential of machine learning assisted image analysis for medical diagnostics and research applications.

7.2 Demonstration

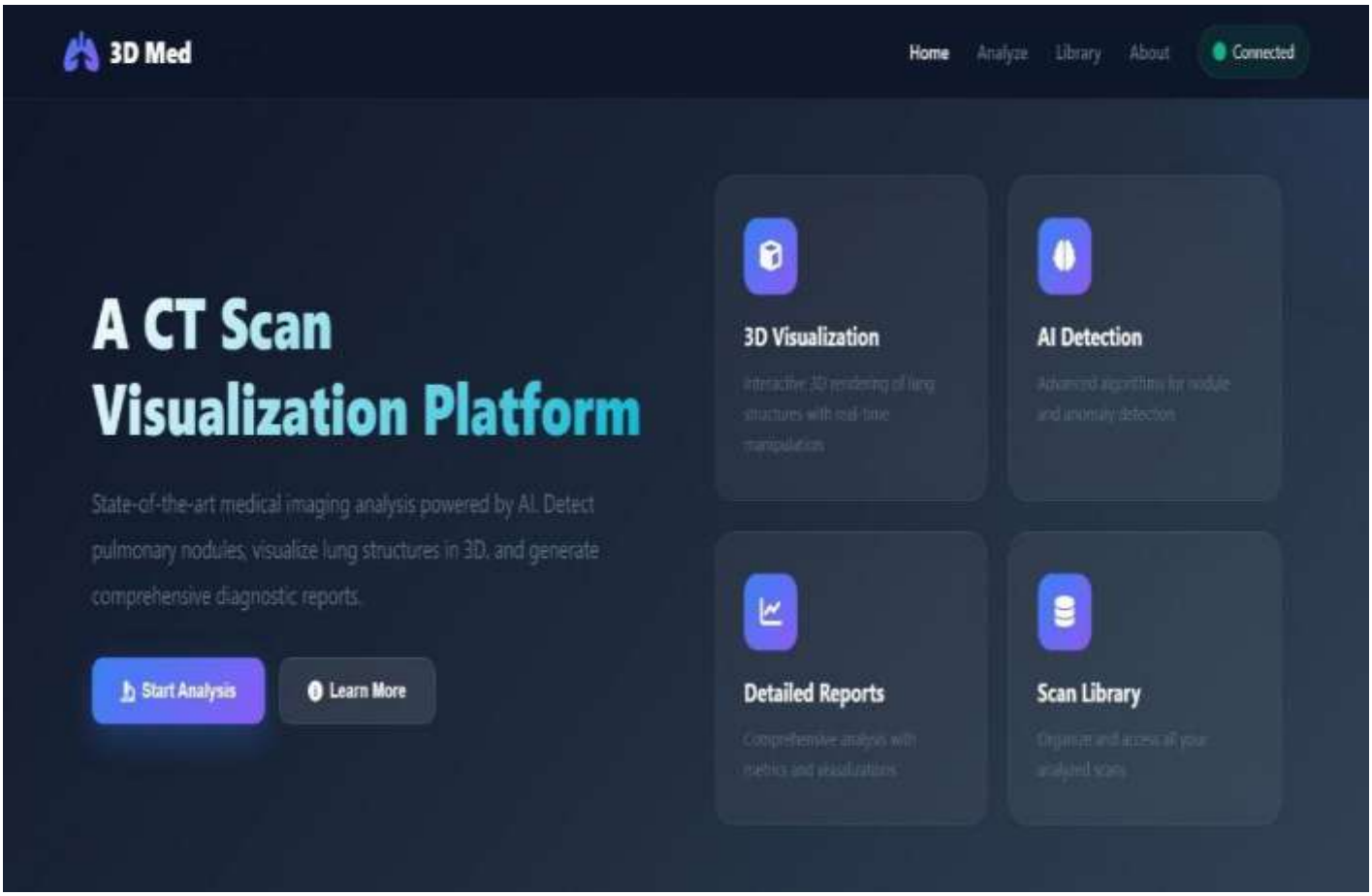

figure 7.2.1 HOME PAGE

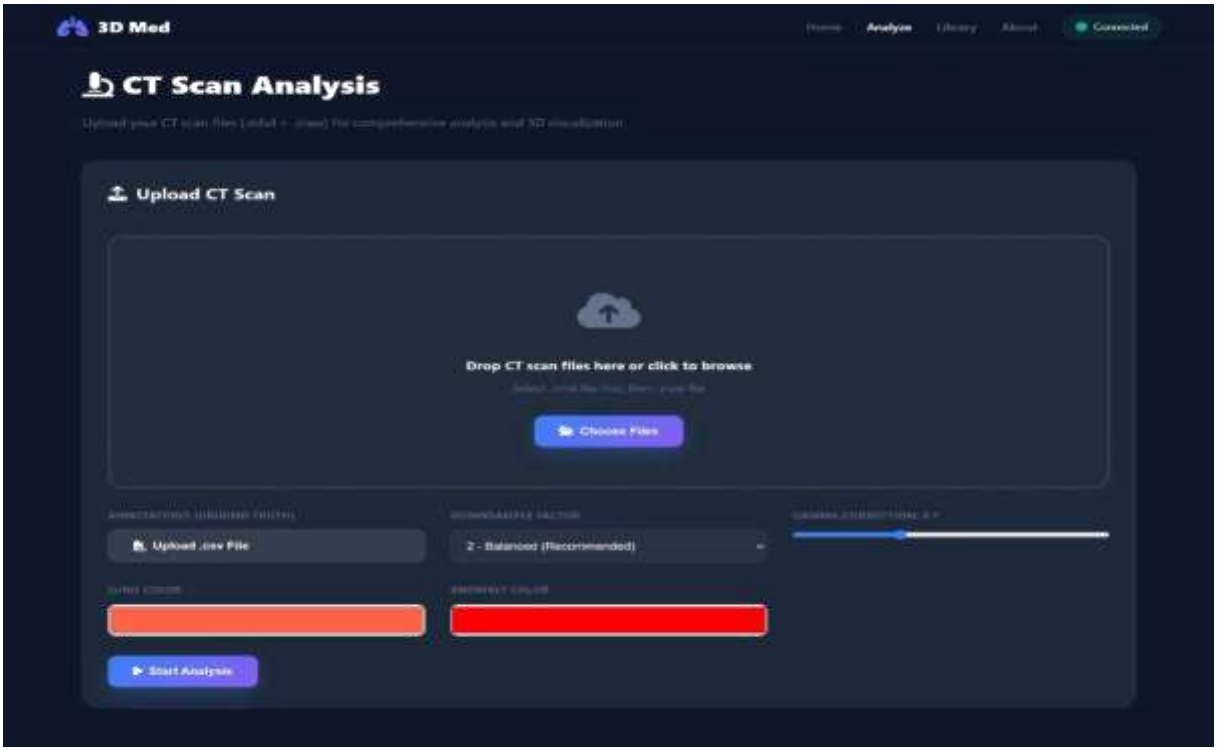

figure 7.2.2 ANALYSIS PAGE

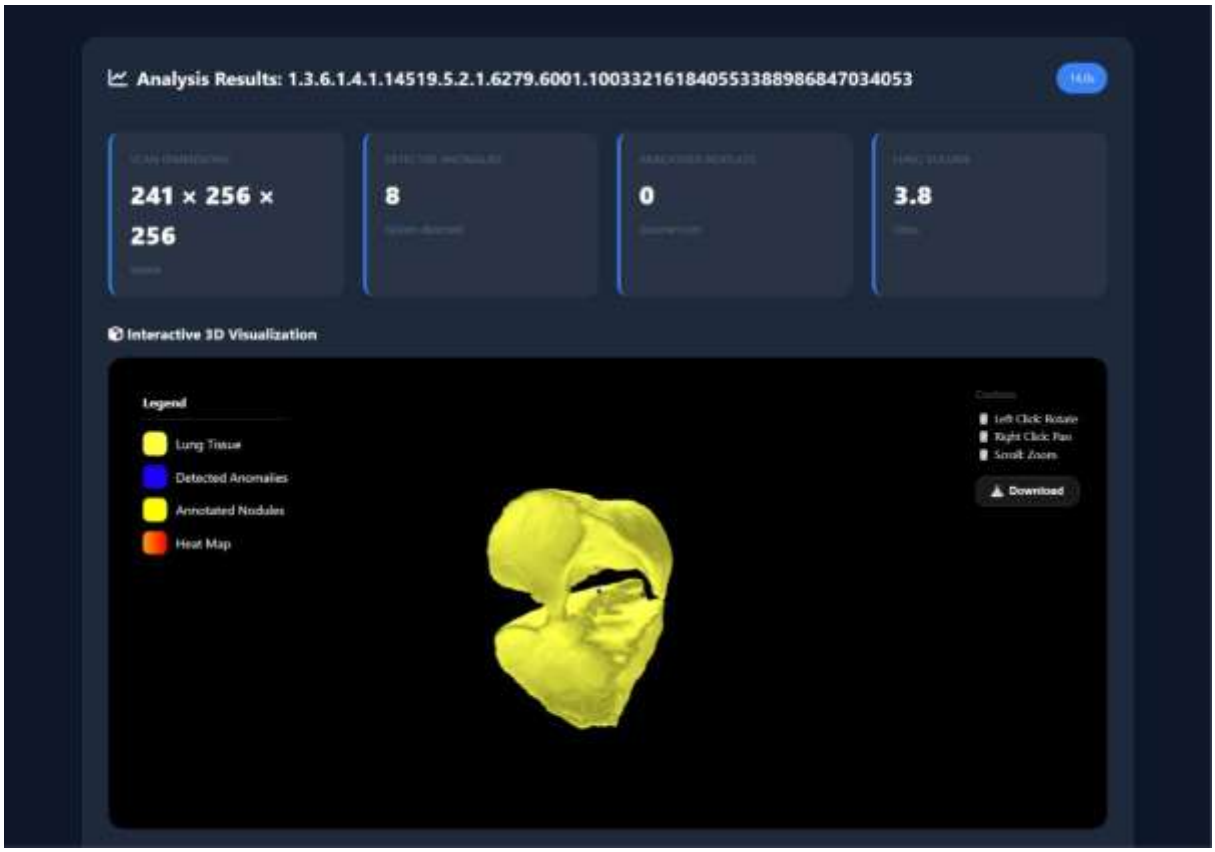

fig 7.2.3 ANALYSIS PAGE(3D MODEL LUNGS)

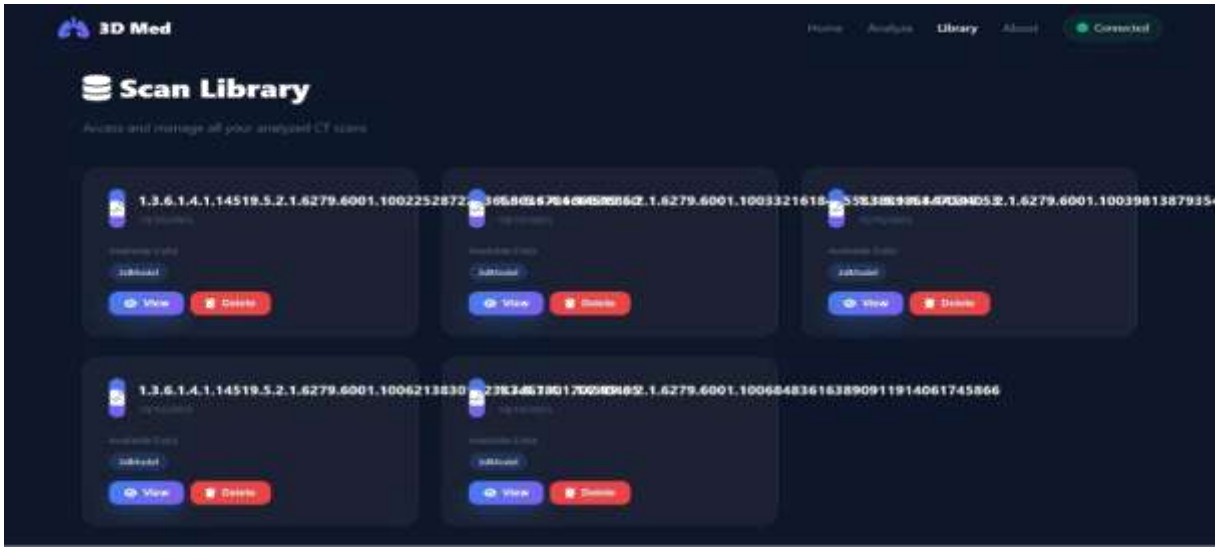

Figure 7.2.4 SCAN LIBRARY PAGE

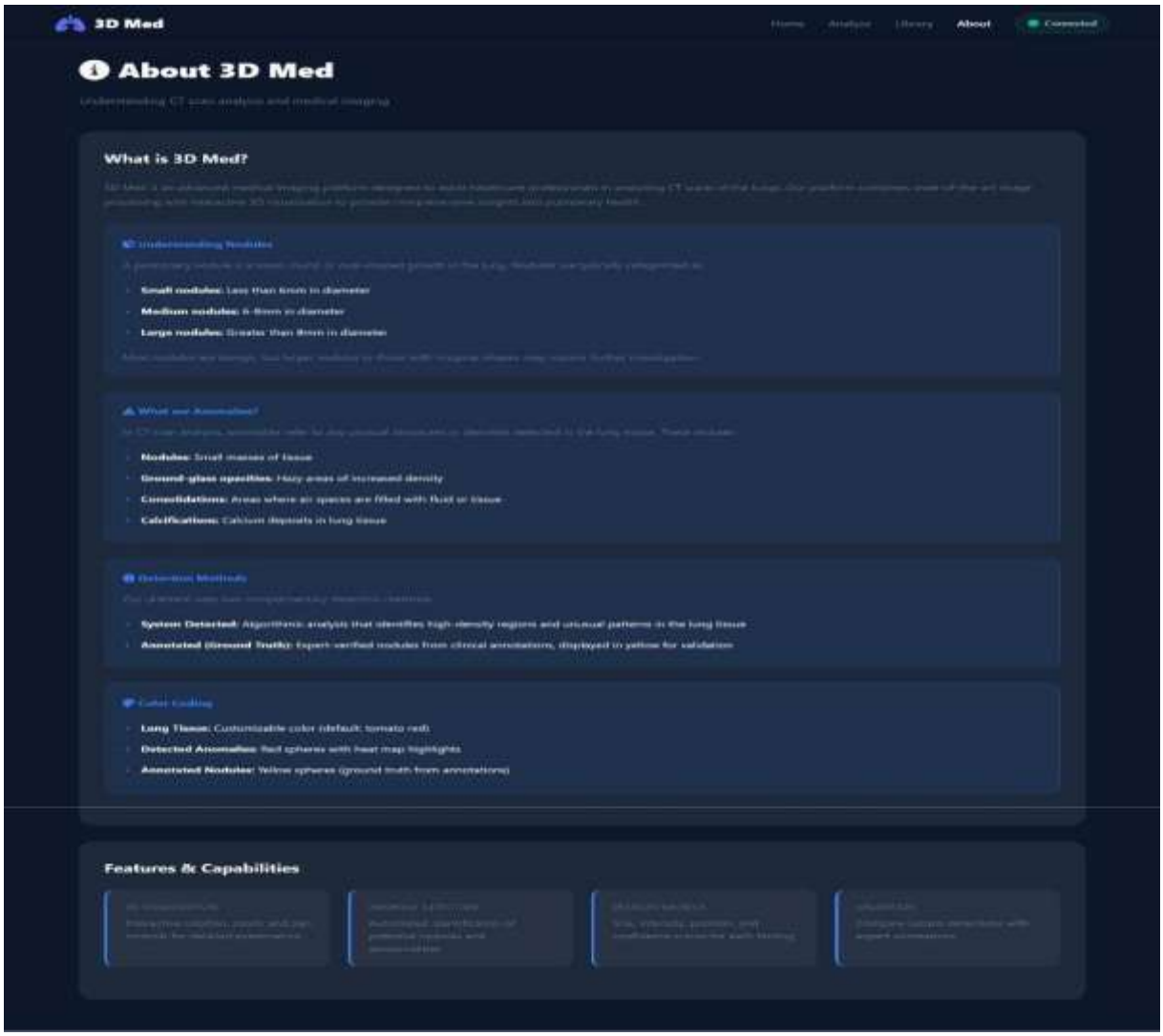

Figure 7.2.5 ABOUT PAGE

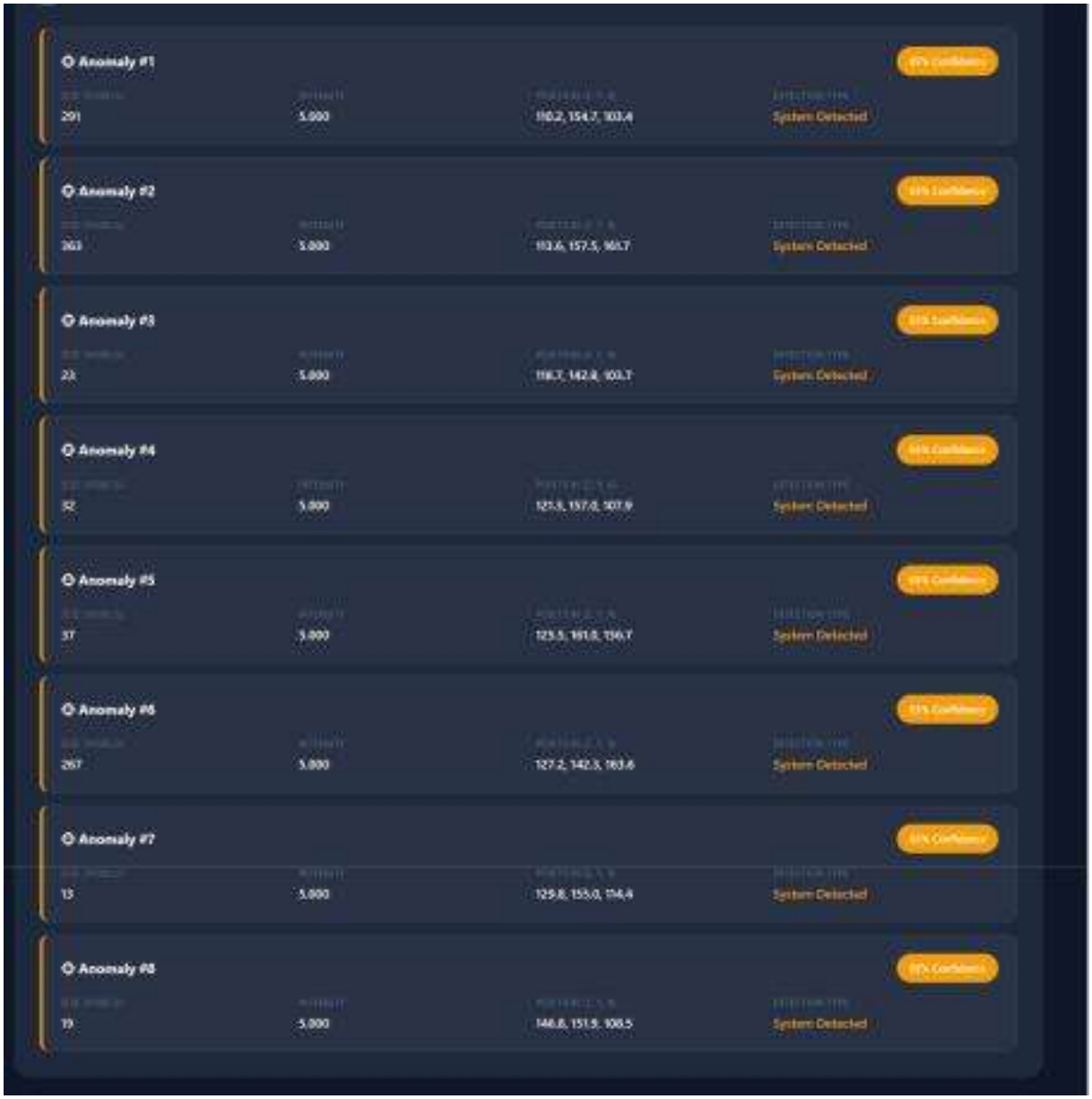

Figure 7.2.6 ANAMOLY PAGE

Chapter 8

Conclusion & Future Scope

8.1 Conclusion

The proposed system, 3D Med, demonstrates the potential of combining deep learning–based CT scan analysis with interactive 3D visualization to support medical diagnosis and education. Unlike conventional approaches that primarily provide classification outputs, this system emphasizes interpretability and usability by highlighting anomalies, generating quantitative metrics, and presenting results through an interactive web-based interface. The architecture integrates preprocessing, anomaly detection, and visualization into a seamless workflow, ensuring that raw CT scans are transformed into actionable insights that are both accurate and visually explainable.

Through its modular design, 3D Med addresses key limitations identified in existing systems, such as the black-box nature of AI models, limited visualization capabilities, and lack of user centered tools. By leveraging 3D rendering, GLTF-based visualization, and web accessibility, the system not only assists radiologists in making faster, evidence-based decisions but also provides a valuable platform for medical students to explore CT anatomy and disease patterns.

The use of widely available technologies such as Flask, SimpleITK, and Three.js ensures scalability, portability, and adaptability to diverse healthcare and educational environments.

In conclusion, 3D Med serves as both a diagnostic assistant and an educational tool, bridging the gap between artificial intelligence and clinical practice. While the current implementation lays a strong foundation, future enhancements such as integration with hospital PACS systems, support for multi-disease classification, and incorporation of uncertainty estimation will further increase its clinical relevance and reliability. With its focus on accuracy, explainability, and visualization, 3D Med represents a step toward the next generation of AI-powered medical imaging solutions.

8.2 Future Scope

Although the system performs efficiently in its current form, there are several areas for enhancement and expansion to make it more comprehensive, clinically relevant, and intelligent.

1. **Multi-Disease Support:** Extend the analysis pipeline to detect and classify multiple lung diseases such as cancer, tuberculosis, pneumonia, and fibrosis alongside COVID-like infections.

2. **Advanced Explainability:** Incorporate transformer-based vision models (e.g., Vision Transformers, MedCLIP) and advanced explainability tools to provide more reliable and clinically interpretable heatmaps.

3. **Integration with Hospital PACS:** Connect the system with existing hospital Picture Archiving and Communication Systems (PACS) to allow seamless real-world deployment and access to patient data directly from hospital workflows.

4. **Cloud-Based Deployment:** Deploy the system on cloud platforms (AWS, GCP, Azure) to handle large datasets and allow remote accessibility for multiple hospitals, research labs, or medical institutes simultaneously.

5. **Uncertainty Estimation:** Add probabilistic models or ensemble approaches to provide confidence scores, helping clinicians identify cases that require further manual review.

6. **Longitudinal Analysis:** Enable tracking of disease progression by comparing multiple CT scans of the same patient over time, providing insights into treatment effectiveness.

7. **Automated Reporting:** Generate structured diagnostic reports summarizing detected anomalies, volumetric measurements, and severity indices, reducing manual reporting effort for radiologists.

8. **Educational Enhancements:** Integrate guided learning modules where medical students can explore CT scans, toggle anomaly overlays, and receive AI-based annotations for training purposes.

In conclusion, this project lays the foundation for AI-powered medical visualization systems by combining automated CT analysis with interactive 3D exploration. With future improvements in scalability, explainability, and clinical integration, 3D Med has the potential to evolve into a comprehensive diagnostic assistant and educational platform, bridging the gap between artificial intelligence, medical practice, and healthcare learning.

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